MIL-C-23092B(SHIPS)
26 February 1976
SUPERSEDING
MIL-C-23092A(SHIPS)
1 November 1968
(See 6.3)

MILITARY SPECIFICATION

CEMENT, NATURAL RUBBER; MAGNETIC MINESWEEPING CABLE REPAIR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a natural rubber cement for making vulcanized and unvulcanized bonds of natural or synthetic rubber to vulcanized synthetic rubber for the repair of magnetic minesweeping cable.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on data of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-566 - Boxes, Folding Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

MILITARY
MIL-T-22755 - Tape, Repair, Magnetic Minesweeping Cable Sheath.

STANDARDS

FEDERAL
FED-STD-601 - Rubber, Sampling and Testing.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC., AGENT
National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street N. W., Washington, D. C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material The cement shall consist of natural rubber dissolved in a solvent. No reclaimed rubber shall be used.

3.2 Solids content. The solids content of the cement shall be not less than 15 percent, by weight, when tested in accordance with 4.4.1.

3.3 Viscosity. The cement shall have a viscosity of not more than 150 seconds nor less than 110 seconds when tested in accordance with 4.4.2.

3.4 Adhesion.

3.4.1 Initial adhesion The adhesion of the cement in both vulcanized and unvulcanized repairs shall be not less than 15 pounds per inch (lb/in) when tested in accordance with 4.4.3.

3.4.2 Aged adhesion. After the cement has been aged as specified in 4.4.4 and then tested in accordance with 4.4.3, the adhesion of the cement in both vulcanized and unvulcanized repairs shall be not less than 15 lb/in.

3.5 Stability. The cement shall not gel or otherwise deteriorate when aged as specified in 4.4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4 2.1 Lot For purposes of inspection, a lot shall consist of all cement from one production batch offered for delivery at one time.

4.2.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level of 2 5 percent defective for the examination specified in 4.3.1.

4.2.3 Sampling for lot acceptance tests. From each lot the inspector shall take enough material to provide two one pint samples for the lot acceptance tests specified in 4.3.2.

4.3 Examination and tests.

4.3.1 Examination. Each filled container in the sample specified in 4.2.2 shall be examined to verify compliance with this specification. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.2 Lot acceptance tests. Each of the samples selected in accordance with 4.2.3 shall be subjected to all the tests specified in 4.4 If any one of the samples fails to conform to any of the test requirements, the lot which it represents shall be rejected.

4.4 Test procedures.

4.4.1 Solids content. The solids content of two specimens of the cement shall be determined as described in 4.4.1.1 through 4.4.1.3.

4.4.1.1 Specimens. Each specimen shall consist of approximately 10 grams of thoroughly stirred cement.

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4.4.1.2 Procedure. Each specimen shall be tested in accordance with the following procedure:

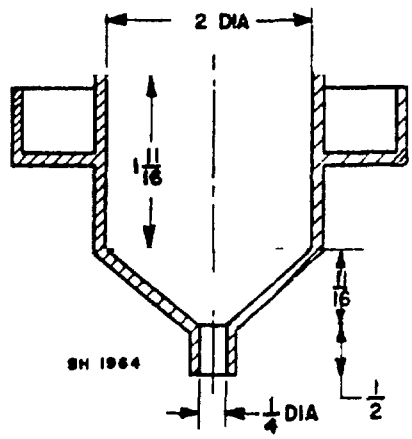
(a) A suitable container and cover shall be weighed.
(b) The specimen shall be poured into the container and covered.
(c) The specimen, container, and cover shall be weighed.
(d) The cover shall be removed from the container.
(e) The container and specimen shall be heated to 70°C ± 1°C and maintained at that temperature until their weight becomes constant.
(f) The specimen, container, and cover shall be cooled in a desiccator and weighed.

4.4.1.3 Calculations. The solids content shall be expressed as the percentage of nonvolatile matter present in the specimen. It shall be calculated as follows:

$$\text{Solids content, percent} = \frac{\text{weight of residue}}{\text{original weight of specimen}} \times 100$$

4.4.2 Viscosity. The viscosity of two specimens of the cement shall be determined as described in 4.4.2.1 through 4.4.2.3.

4.4.2.1 Specimens. Each specimen shall consist of enough cement to fill the cup shown in figure 1.



All dimensions in inches

Figure 1 - Schematic sketch of viscosity cup.

4.4.2.2 Apparatus. The cup shown in figure 1 (commonly called a Ford cup) shall be used in this test.

4.4.2.3 Procedure. Both the specimen of cement to be tested and the apparatus shall be adjusted to the testing temperature (23°C ± 1°C) before starting the test. Stir the cement thoroughly, allowing a minimum of solvent to evaporate. Level the apparatus and then place one finger over the orifice of the cup. Fill the cup with cement rapidly and allow a moment or two for air bubbles to rise. Wipe the finger away from the orifice of the cup and simultaneously start a stopwatch. Observe the stream of cement flowing from the orifice of the cup and stop the watch at the first break in the stream. The time recorded by the stopwatch, in seconds, shall be considered the viscosity of the specimen.

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4.4.3 Adhesion.

4.4.3.1 Materials. Determinations of adhesion require the use of rubber tape conforming to type I of MIL-T-22755 and of two reference standards. The reference standards shall be prepared according to the following formulas, using the materials shown or their equals.

Material	Parts by weight Standard No. 1	Standard No. 2
SBR 1500	100	---
Neoprene GN-A	---	100
Wyex (Easy processing channel black)	40	27
Protox 166 (zinc oxide treated with proprionic acid)	5	5
Stearic acid	1.5	0.5
Altax (Di-2 benzothiazyl disulfide)	3	---
Sulfur	2	---
Neozone A (phenyl alpha naphthylamine)	---	2
Sodium acetate	---	1
Circo light process oil (light naphthenic petroleum product)	---	3
Maglite D (magnesium oxide)	---	4
Vulcanize	50 minutes at 144°C ± 1°C	30 minutes at 142°C ± 1°C

4.4.3.2 Vulcanization of standards. The standards shall be vulcanized in a press under a pressure of not less than 500 pounds per square inch (lb/in^2). The vulcanization shall be for the length of time and at the temperature specified in 4.4.3.1. The vulcanized standards shall be 0.125 inch plus or minus 0.010 inch thick and of sufficient size to permit trimming to the dimensions of 4.4.3.2.

4.4.3.3 Preparation of adhesion specimens. The vulcanized sheets of standards No. 1 and 2 shall be cut into strips 2 inches plus or minus 0.010 inch wide by 6 inches plus or minus 0.010 inch long. These strips shall be roughened on one face by buffing on a course grinding wheel or other suitable means. Buffing dust shall be removed by means of a clean brush. One brush coat of the rubber cement being tested shall be applied to the roughened surface and allowed to dry until the odor of solvent is no longer apparent. A piece of the separator from the tape conforming to type I of MIL-T-22755 shall be applied to the cemented surface to cover about 2 inches of the length at one end of the strip.

4.4.3.3.1 Procedure for vulcanized repairs. One or more 6- inch lengths of tape conforming to type I of MIL-T-22755 shall be rolled down against the cemented surface prepared as specified in 4.4.3.3 and over the separator, until the thickness of the assembly is 0.260 inch plus or minus 0.010 inch. The assembly shall then be placed in a 2-inch by 6-inch by 0.250-inch mold and vulcanized for 30 minutes at 154°C ± 1°C under a pressure of not less than 500 lb/in^2. The vulcanized assembly shall then be reduced to a width of 1 inch plus or minus 0.010 inch by cutting off both sides. The adhesion shall then be determined as specified in 4.4.3.4.

4.4.3.3.2 Procedure for unvulcanized repairs. A 6-inch length of tape shall be placed on the cemented surface prepared as specified in 4.4.3.3 and over the separator. The roller described in 4.4.3.3.2.1 shall then be passed over the tape specimen, once in each direction at a rate of approximately 2 inches per second, without application of pressure other than that due to the weight of the roller. After the tape specimen has remained on the strip for 30 minutes plus or minus 2 minutes, its adhesion shall be determined as specified in 4.4.3.4.

4.4.3.3.2.1 Roller. The roller required in 4.4.3.3.2 shall be not less than 3 inches in diameter and not less than 2 inches wide. The weight of the roller shall be 5 pounds plus or minus 0.1 pound.

4.4.3.4 Determination of adhesion. The specimens, prepared as specified in 4.4.3.3 and 4.4.3.3.1 or 4.4.3.3.2, shall be subjected to the adhesion test specified in method 8011 of FED-STD-601. The 2 inches of tape lying over the separator of each specimen being tested shall be peeled back by hand and clamped in the upper jaw of the testing machine. The exposed 2 inches of base standard strip with separator shall be attached to the lower jaw of the testing machine and the adhesion determined.

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4.4.4 Oven aging. Oven aging shall be conducted in an apparatus of the type specified in method 7221 of FED-STD-601. The oven used shall also be equipped with an explosion-release safety hatch. The cement shall be aged for 28 days at 60°C ± 1°C. An alternate aging device is a water bath maintained at 60°C ± 1°C in which the cement container shall be completely immersed. The cement shall be aged in a tightly closed container used in filling the contract or order. At the end of the aging period, the cement shall be removed from the oven and cooled for 24 hours plus or minus 1 hour at 22°C ± 1°C. The container shall be opened and the cement shall be examined for gelling or any other sign of deterioration. The adhesion tests required in 3.4.2 shall be performed using this aged cement.

4.4.5 Test procedures and test reports. The supplier shall prepare test procedures and test reports in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.5 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2). The cement shall be furnished in glass containers of 4 or 8-ounce capacity or in collapsible tubes of 4 or 8-ounce capacity as specified (see 6.2).

5.1.1 Level A. The cement, furnished in the quantity specified (see 6.2), shall be packaged in containers as follows:

(a) Glass containers. The glass container shall be of a light-resistant glass with plastic screw-caps and liner type closure. With the liner in place, the screw-cap shall not crack, split or break when applied to the container and tightened. Closure shall be tightened to the degree required to prevent loosening of the cap or loss of contents. The cap closures shall be provided with cellulose hydrate bands or equivalent as an outer seal. Outer seals shall cover the cap externally, extending well down on the neck of the glass container and slightly over the top edge of the cap. When dried, the band shall conform to the contour of the cap and neck of the bottle. Twelve glass containers of the same capacity shall be packaged in a fiberboard box of minimum 200 pounds bursting strength conforming to PPP-B-636 class domestic. Carton design shall include partitions, liners, and top and bottom pads. Arrangement shall be at the suppliers option.

(b) Collapsible tubes. The collapsible tubes shall consist of a seamless, cylindrical body (wall) with integral shoulder and threaded neck. Tubes shall be fabricated of tin-copper alloy, tin-lead alloy, or aluminum and shall have suitable flexible characteristics and an opening which shall permit ready dispensing of the material. Tube walls shall be of uniform thickness and may be treated with an interior protective coating. An inner seal in the form of a dish shall be adhered completely to the mouth of the tube. The tubes shall be provided with a screw cap to protect the contents once the inner seal is punctured. Each tube shall be further packaged in a paperboard box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Twelve packaged tubes of the same capacity shall be placed in a paperboard or fiberboard box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636 respectively. Closure of boxes shall be in accordance with the applicable box specification or appendix thereto.

5.1.2 Level C. The cement furnished in the quantity specified (see 6.2), shall be packaged in glass containers or collapsible tubes as specified in 5.1.1 and overpacked in paperboard or fiberboard containers in a manner to afford protection against physical damage, loss of contents, or breakage during shipment from the supply source to the first receiving activity for immediate use. The suppliers normal packaging methods may be utilized when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Rubber cement, packaged as specified in 5.1, shall be packed in fiberboard boxes conforming to PPP-B-636, class weather-resistant, Grade V with variety and style at the suppliers option. Box closure, waterproofing, and reinforcing shall be as specified for method V in accordance with the appendix to the box specification.

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5.2.2 Level B. Rubber cement, packaged as specified in 5.1, shall be packed in fiberboard boxes conforming to PPP-B-636 class domestic. Closure shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Rubber cement, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight and National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2), or herein, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include type I Shelf-life markings. Data manufactured and expiration dates shall be included.

5.3.1 Precautionary markings. Each interior package and exterior shipping container shall be marked on two opposite sides as follows:

"STORE IN A COOL DRY PLACE"

Containers of glass bottles shall be marked with a fragile label and stenciled denoting "GLASS, HANDLE WITH CARE".

6. NOTES

6.1 Intended use. Natural rubber cement is used for making vulcanized and unvulcanized bonds of natural or synthetic rubber to vulcanized synthetic rubber for the repair of magnetic minesweeping cable.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Size container (see 5.1).
(c) Levels of packaging and packing required (see 5.1 and 5.2).
(d) Special marking required (see 5.3).

6.2.2 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

	Specification paragraph	Data requirements	Service	Applicable DID	Options
(a)	4.4.5	Procedures, test	SH	UDI-T-23732	---
(b)	4.4.5	Report, test/ inspection	SH	UDI-T-23473	---

(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.3 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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